

Task 1 — Introduction to Cyber Security

Internship: Cyber Security Internship

Organization: Next Gen Software Hub Pvt. Ltd.

Week: 1

Task: Introduction to Cyber Security

1. Introduction

Cyber security is the practice of protecting computer systems, networks, applications, and data from unauthorized access, attacks, damage, or disruption.

Cyber security is important because individuals and organizations store and process large amounts of sensitive information digitally. Security measures help protect this information and ensure that computer systems remain reliable and available.

2. Cyber Security Threats and Attacks

A **cyber threat** is a potential danger that can affect a computer system, network, or data.

A **cyber attack** is an actual attempt to exploit a weakness in a system.

Common examples include malware, phishing, and ransomware.

3. CIA Triad

The CIA Triad represents three important principles of cyber security:

Confidentiality

Confidentiality ensures that information is accessible only to authorized people.

Example: Protecting personal information using passwords and access controls.

Integrity

Integrity ensures that information remains accurate and is not modified without authorization.

Example: Preventing unauthorized modification of important records.

Availability

Availability ensures that systems and information are accessible when required.

Example: Keeping an organization's website or server available to authorized users.

Therefore:

C — Confidentiality → Protect information

I — Integrity → Keep information accurate

A — Availability → Keep systems accessible

4. Common Cyber Threats

4.1 Malware

Malware is malicious software designed to harm, disrupt, or gain unauthorized access to a computer system.

Examples include:

- Virus
- Worm
- Trojan
- Spyware

4.2 Phishing

Phishing is a technique in which attackers use fake emails, messages, or websites to trick users into providing sensitive information.

Example: A fake banking message asking a user to click a link and enter their username and password.

4.3 Ransomware

Ransomware is a type of malware that locks or encrypts a victim's data and demands payment to restore access.

Example: A user's files become inaccessible and a message demands payment to decrypt them.

5. Practical Threat Identification

Scenario	Type of Threat
A malicious program infects a computer	Malware
A fake email asks for your banking password	Phishing
Files are encrypted and money is demanded to unlock them	Ransomware

6. Conclusion

Through this task, I learned the fundamentals of cyber security, including **cyber threats, cyber attacks, and the CIA Triad**. I also learned to identify common threats such as **malware, phishing, and ransomware**.